

Design Space Analysis and ρ – η Pareto Optimization of LC Output Filters for Switch-Mode AC Power Sources

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Abstract—In this study, the design space (DS) concept is refined and utilized to design the output filter of a 10-kW, four-quadrant, three-phase, switch-mode controllable AC power source (CPS). The DS concept is based on the simultaneous consideration of multiple criteria that are derived from application-oriented specifications for the CPS regarding quality and transient response of the output voltage, and limited reactive power demand of the output filter. In this paper, the output filter of the studied CPS needs to satisfy six different criteria, which, in the case of a single-stage LC filter, leads to bounds on the values of L and C that can be indicated by boundary curves in an L – C plane. The intersection set of all boundary curves defines the DS, in which all six specifications can be fulfilled. For the considered requirements, it is shown that the DS is empty for a single-stage LC filter, but exists for a two-stage LC filter, which is therefore employed as an output filter of the CPS. To fully exploit the 4D DS of the two-stage LC filter, a multi objective optimization, resulting in the power density–efficiency Pareto front, is performed to determine the most compact and/or most efficient filter design among all possible filter realizations with parameters in the DS. From the outcome of this optimization, the filter design with the highest power density of 14.6 kW/dm³ (239 W/in³), for an efficiency of 99.4%, is realized in hardware. Finally, all six specifications for the CPS are successfully verified by experimental measurements.

Index Terms—Converter control, design space, LC output filter, Pareto optimization, power source.

I. INTRODUCTION

FOR developing and testing power electronics equipment and associated control strategies, controllable AC power sources (CPSs), which have the capability to emulate certain electrical characteristics, feature shorted durations of development and test procedures, i.e., are saving costs and efforts. Example applications of CPSs are motor emulators for testing inverters [1]–[3] (also in combination with power hardware-in-the-loop simulations [4]); grid emulators for testing utility connected distributed generators [5]–[7], such as fuel cell based systems [8] (this can include the testing of safety and protection functions [9] and testing of the implemented control strategy, which again can be done with hardware-in-the-loop simulations

[10]); and grid emulators for optimizing the control scheme for traction vehicles [11].

Such CPSs can basically be implemented with a linear (e.g., [12]) or a switch-mode power amplifier (e.g., [13]). To reduce the weight and the volume, a switched system is preferred.

The output stage of the CPS considered in this paper is shown in Fig. 1. As further explained in Section II, the system needs to comply with specifications regarding qualities and transient responses of all three output voltages $v_{A,out}$, $v_{B,out}$, and $v_{C,out}$ and needs to achieve the efficiency specified at the nominal operating point (cf., Table I). Because the CPS actually is a switch-mode power supply, the amplitudes of the harmonics in the switched output voltages (e.g., v_{A0} in Fig. 1) need to be attenuated by an output filter to comply with the given requirements. In the simplest case, the output filter is implemented as a single-stage LC filter according to Fig. 2(a). The component values used for the LC filter are restricted, not only in their minimum values, to fulfill the output voltage quality requirements, but also in their maximum values, to satisfy the dynamic specifications.

A literature survey reveals numerous publications on the design of the output filter with boundary conditions defined by specifications, which typically deal with only one aspect, such as conducted EMI [17]–[19], output voltage ripple [20], [21], or current ripple [22]–[24]. For the case at hand, however, multiple requirements need to be satisfied. For this purpose, the concept of the design space (DS), first employed in [25] to design a single-stage LC filter for a 20-W synchronous buck converter in order to comply with static and dynamic output voltage regulation specifications, is considered most promising. The DS concept is based on multiple criteria that are derived from all dynamic and static specifications. In case of [25], these criteria define bounds on the values of L and C , which can be represented by boundary curves in an L – C plane. The intersection set of all boundary curves builds the DS. Thus, the strength of the DS approach is that all combinations of filter parameter values that comply with all specifications are identified. Another publication related to the DS concept details the implications of changing specifications and/or requirements on the DS calculated for a 0.6-W buck converter including PWM delays and filter resonances in voltage mode [26]. Both studies [25], [26] are limited to single-stage LC filters and do not detail any further optimization of the filter using the calculated DS, e.g., with respect to power density or efficiency.

Subsequent to the identification of the DS, the filter can be optimized, e.g., by means of multiobjective optimization with

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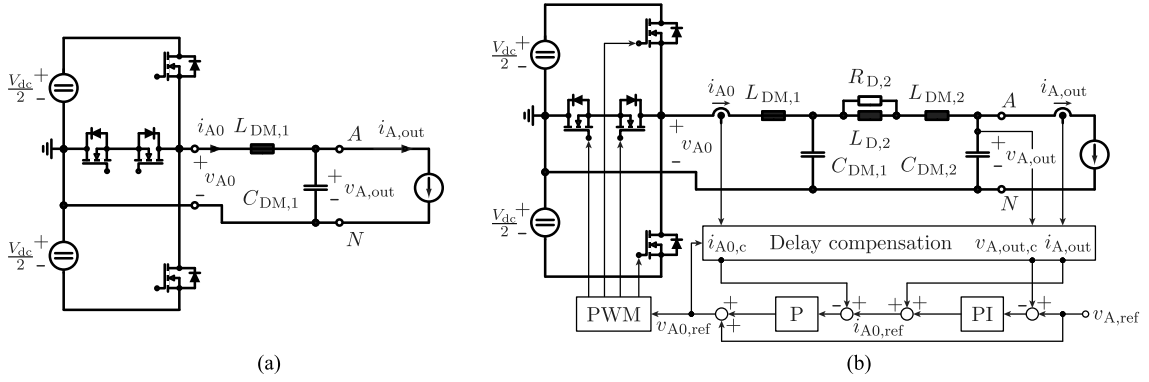


Fig. 2. (a) Simplified single-phase equivalent circuit of the CPS for a single-stage LC output filter and (b) a two-stage LC output filter with a series $R_{D,2} L_{D,2}$ damping branch of the second filter stage. The load is represented as a current source. In (b), the control scheme [43] employed in Section V is added to the equivalent circuit. It consists of an outer voltage control loop for $v_{A,out}$, with feedforward of the reference voltage v_{ref} , an inner current control loop for i_{A0} , with feedforward of the load current $i_{A,out}$, and a sampling plus PWM delay compensation [44] to achieve a satisfactory control performance (cf., Section V).

TABLE II
REQUIRED PROPERTIES OF THE CPS

Output voltage quality	$THD_v < 2.5\%$ (cf., IEEE 1547 [15])
Output voltage slew rate	$SR \geq 203 \text{ V/ms}$
Max. transient output voltage dip $\Delta v_{A,out}$ due to a stepwise output current change $\Delta i_{A,out}$	$\left \frac{\Delta v_{A,out}}{\Delta i_{A,out}} \right \leq \frac{28 \text{ V}}{5 \text{ A}} = 5.6 \Omega$
Limited filter capacitor reactive power demand	$Q_{cap,max} \leq 10\% \cdot \frac{P_{out,n}}{3} = 333 \text{ VA}$
Conducted EMI	CISPR 11, Class A [16]

MOSFETs. Furthermore, only a constant output frequency is considered, since the CPS is used for emulating a 50 Hz grid. Table II lists additional requirements, which are summarized below.

$\emptyset$ requirements are differently specified for sinusoidal and arbitrary output voltage waveforms: For sinusoidal output voltages, a total harmonic distortion of $THD_v < 2.5\%$ (cf., IEEE 1547, IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems [15]) is specified; for arbitrary output voltage waveforms, a minimum average slew rate SR_{min} is defined. In this context, the

$\emptyset$ of $\Delta v_{ref} = 10\% \cdot V_{out,n,peak} = 32.5 \text{ V}$ is required to be completed after $t_{step} = 160 \mu\text{s}$ [cf., Fig. 3(a)]. This requirement is summarized using a minimum average slew rate¹

$$SR_{min} = \frac{10\% \cdot 325 \text{ V}}{160 \mu\text{s}} = 203 \text{ V/ms.} \quad (1)$$

Accordingly, with SR_{min} , the converter can generate a sinusoidal output voltage reference with a frequency of 1 kHz and an amplitude of 32 V.² With this, and according to the results of

the derivation presented in Section III-A, a small-signal bandwidth of the closed output voltage control loop of approximately 4 kHz is estimated, i.e., the CPS is capable of tracking a reference voltage which leads to an output voltage waveform that does not exceed the slew rate specified in (1) and contains frequency components up to 4 kHz.

An output voltage transient provoked by a step change of the voltage reference, v_{ref} , generates also a bridge-leg output current transient, which, depending on the output current, possibly drives the converter in its current limitations. This would then lead to a nonlinear behavior of the controlled converter, which is undesired. Thus, for the slew rate calculation, it is assumed that the CPS can always abruptly increase its output voltage by 32.5 V without running into its current limitation. This is experimentally verified in Section V-A, where the required increase of the bridge-leg output current is roughly 4 A.

The $m m m$ $\emptyset$, $\Delta V_{A,out,max}$, that occurs $\Delta I_{A,out}$ can be characterized by means of an impedance

$$Z_{step,max} = \left| \frac{\Delta V_{A,out,max}}{\Delta I_{A,out}} \right|. \quad (2)$$

According to predefined requirements, $\Delta V_{A,out}$ must not exceed 28 V (8% of $V_{A,out,max,peak}$) for $\Delta I_{A,out} = 5 \text{ A}$, which yields $Z_{step,max} = 5.6 \Omega$ [cf., Fig. 3(b)].

To limit the current stress on the components of the power circuit, the $m m m$ m $\emptyset$

to a required time interval of $\Delta t = t_{step} = 32 \text{ V} / 201 \text{ V/ms} = 159.2 \mu\text{s} \approx 160 \mu\text{s}$.

¹The definition of the average slew rate adopted in this paper is different from the definition of the slew rate known from operational amplifiers (OpAmps). For OpAmps, the slew rate is defined as the “maximum rate at which the output voltage of the OpAmp can change,” i.e., $(dv/dt)_{max}$, which is typically determined for a full-scale large-signal input voltage step [47].

²The $|dv/dt|$ of a sinusoidal function with amplitude V_{sig} and frequency f_{sig} is maximal at its zero crossings and can be calculated as $2 \cdot \pi \cdot f_{sig} \cdot V_{sig}$, which gives 201 V/ms for $V_{sig} = 32 \text{ V}$ and $f_{sig} = 1 \text{ kHz}$. Accordingly, to change the output voltage by 32 V with an average rate of change of 201 V/ms leads

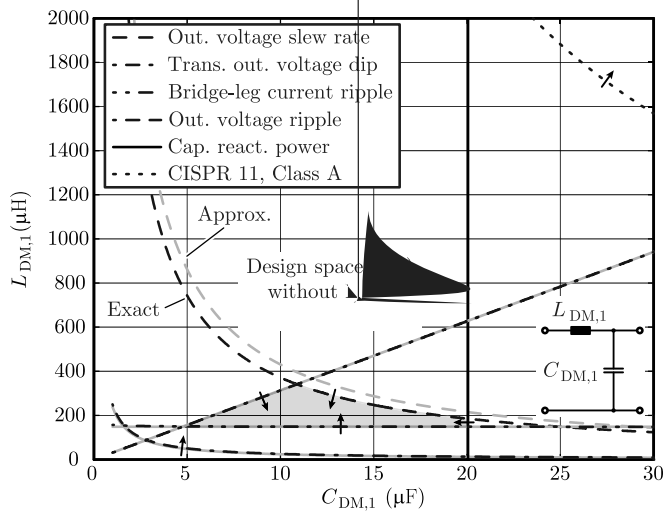
is obtained, such that the maximum possible rate of change of the output voltage is achieved.

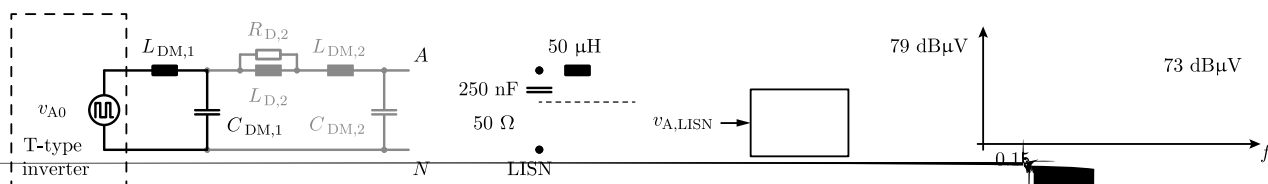
If an input voltage step $\Delta v_{A0} > 0$ is applied to a single-stage LC filter [cf., Figs. 1 and 2(a)], the current through the filter inductor increases and subsequently the capacitor voltage, i.e., output voltage, starts to change as well. The presented result considers the response to a positive input voltage step (the bridge-leg constantly applies $+V_{dc}/2$ until the output voltage reaches its new reference value) and neglects the implication of the changing capacitor voltage $v_{A,out}$ on the inductor current i_{A0} , which, for $m := v_{A,out}/\frac{V_{dc}}{2}$, gives

$$\Delta v_{A,out} \approx \frac{V_{dc}}{4 \cdot L_{DM,1} \cdot C_{DM,1}} \cdot (1 - m) \cdot \Delta t^2 \quad (5)$$

[thin gray line in Fig. 3(a)]. Accordingly, the idealized slew rate is

$$SR_0(\Delta v_{A,out}) = \sqrt{\quad}$$





which all requirements can be simultaneously fulfilled. The determination of n and k is detailed in Section IV-B.

According to [53], the second LC filter stage is passively damped with a serial RL -branch. The optimal value $R_{D,2,opt}$ of the damping resistor $R_{D,2}$ for a given damping inductance $L_{D,2}$ is computed such that the filter input-to-output transfer function leads to the smallest resonance gain A_{rr} , in contrast to [54], where $R_{D,2}$ is calculated to result in the minimal peak of the filter output impedance. For the filter design

$$a = \frac{L_{D,2}}{L_{DM,2}} = 2, \text{ and}$$

$$R_{D,2} = R_{D,2,opt} = \sqrt{\frac{L_{DM,2}}{C_{DM,2}}} \cdot \frac{2 \cdot a}{\sqrt{2 \cdot a^2 + 6 \cdot a + 4}} \quad (25)$$

are used to obtain a maximum A_{rr} of 6 dB. With $A_{rr} = 6$ dB, the second filter stage is of approximate Chebyshev type (with ripple factor $\epsilon = 0.12$ [55]).

IV. TWO-STAGE FILTER DESIGN

The different criteria detailed in Section III constrain the DS of the two-stage LC output filter as, for example, shown in Fig. 6. However, in order to facilitate the realization of the most suitable output filter for the CPS, within the DS, an appropriate selection procedure has to be defined. In this study, a multi-objective filter optimization, based on the ρ - η PF [35], is conducted for the two-stage LC output filter depicted in Fig. 2(b). The loss and volume models needed for this optimization are summarized in Section IV-A. Section IV-B details the optimization procedure and discusses the calculated optimization results.

A. ρ m

The inductor losses comprise copper and core losses. The calculation of the copper losses considers DC and AC losses (by reason of skin and proximity effects), which, for $L_{DM,1}$, are computed with

$$P_{cu} = R_{dc} \cdot I_{A,out,n}^2 + \sum_{m=1}^{I_m \geq 5\% \cdot I_1} R_{ac,m} \cdot I_m^2 \quad (26)$$

where I_m is the rms value of the m th harmonic of the inductor current ripple. R_{dc} and $R_{ac,m}$ are calculated according to [56], [57] for a temperature of 100 °C. Equation (26) considers all harmonics with an rms value larger than or equal to 5% of the rms value I_1 of the first harmonic. The calculations of the current ripples for the inductors of the second filter stage [cf., Fig. 2(b)] are based on the simplifying assumption that the high-frequency AC components of the current through $L_{DM,1}$, $i_{A0,ripple}(t)$, can be represented by a single harmonic at the switching frequency with an rms value of

$$i_{A0,ripple,rms} = \sqrt{\frac{1}{T_s} \cdot \int_0^{T_s} i_{A0,ripple}(t)^2 \cdot dt}$$

$$\stackrel{m=0.5}{=} \frac{V_{dc,max}}{\sqrt{3} \cdot 16 \cdot L_{DM,1} \cdot f_s} \quad (27)$$

The current ripples through $L_{DM,2}$ and $L_{D,2}$ are then computed based on $i_{A0,ripple,rms}$ and employing linear network analysis.

To realize the winding of the inductors, enameled copper wires are considered, because for the case at hand, enameled wire leads to lower ohmic losses compared to litz wire due to the greater achievable winding area filling factor [17].

The core losses P_{fe} are calculated with the improved generalized Steinmetz equation (iGSE) according to [57], [58]. The Steinmetz parameters k , α , and β , needed for evaluating the iGSE, are given at a core temperature of 100 °C in [59] and [60] for the selected ferrite (N27 and N87 materials) E-cores from TDK EPCOS [61] to realize the inductors. Ferrite cores are utilized because of lower cost compared to nanocrystalline cores and due to a constant inductance over current compared to powder cores.

The evaluation of the copper and core losses requires the bridge-leg output current, $i_{A0,ripple}(t)$, and the flux density in the core, $B(t)$, which are computed for the following worst case conditions:

$$\begin{aligned} \text{max. DC-link voltage: } & V_{dc} = V_{dc,max} = 800 \text{ V}, \\ \text{DC output current: } & I_{A,out} = 17 \text{ A}, \\ \text{modulation index: } & m = v_{A,out} / \frac{V_{dc,max}}{2} = 0.5, \\ \rightarrow \text{DC out. voltage: } & v_{A,out} = 200 \text{ V}. \end{aligned} \quad (28)$$

The modulation index $m = 0.5$ leads to the maximum current ripple ΔI_{A0} (cf., Section III-C) and thus also to the maximum flux density ripple ΔB , which results in the maximum core losses according to the iGSE. To compute the current ripple ΔI_{A0} through $L_{DM,1}$, the voltage across $C_{DM,1}$ is assumed constant over one switching period T_s . The current is increased from the nominal value $I_{A,out,n} = 14.5$ A to $I_{A,out} = 17$ A for the following reason. One application of the CPS could be the emulation of a low voltage mains (e.g., 230 V/400 V). The tolerance of the mains voltage is according to [62] (IEC standard voltages) +10%/−14%. Thus, to supply a load with $P_{A,out,n} = 3.3$ kW for an emulated rms mains voltage of 230 V $\cdot (1 - 0.14) = 198$ V leads to a rms current of $I_{A,out} = 16.7$ A ≈ 17 A.

Finally, to verify that the calculated winding losses P_{cu} and core losses P_{fe} do not lead to excessive temperature rises in the inductor, the maximum temperature $T_{L,max}$ of the inductor is approximated with

$$T_{L,max} = R_{th} \cdot (P_{fe} + P_{cu}) + 40 \text{ °C} \quad (29)$$

assuming an ambient temperature of 40 °C. The thermal resistance R_{th} is calculated with the empirical relation given in [63, Fig. 2] for E-cores

$$R_{th} = \left(53 \cdot (V_{eff} [\text{cm}^3])^{-0.54} \right) \frac{\text{K}}{\text{W}} \quad (30)$$

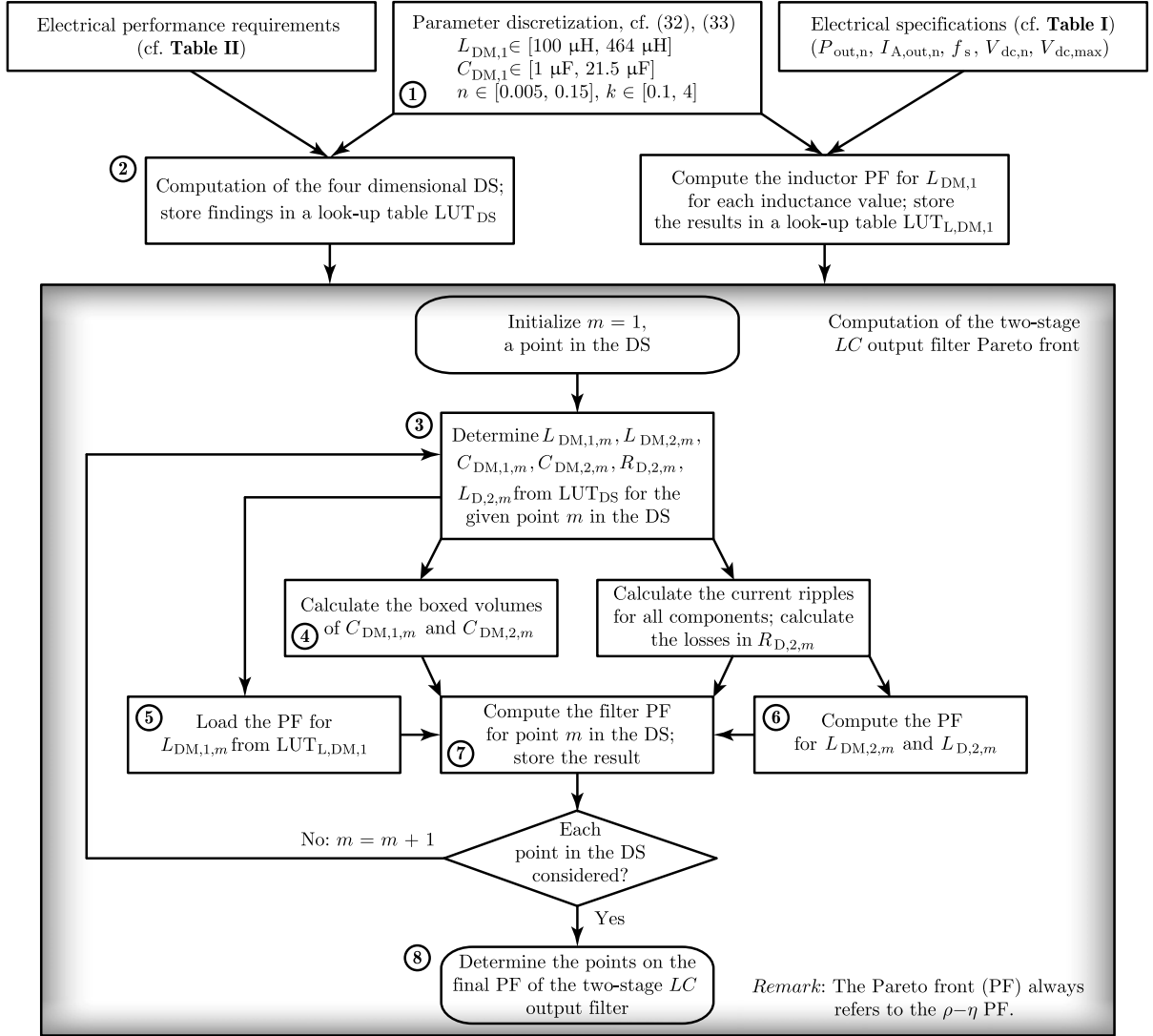


Fig. 7. Flowchart to compute the ρ - η PF for a two-stage LC output filter [cf., Fig. 2(b)].

where V_{eff} is the effective core volume.⁶ Inductor designs which exceed $T_{L,\text{max}} = 100^\circ\text{C}$ are excluded from the Pareto optimization explained in the next section.

The boxed volume of the inductor is calculated with the dimensions of the core and the estimated dimensions of the winding heads (including the bobbin dimensions) [61].

The capacitors of the output filter are realized with X2/305 V rated MKP (polypropylene) film capacitors from EPCOS [64]. Their dielectric losses are negligible compared to the inductor losses. The boxed volume of the capacitor with capacitance $C_{\text{DM},i}$ ($i = 1, 2$) is estimated with

$$V_{C_{\text{DM},i},\text{boxed}} = 4.15 \frac{\text{cm}^3}{\mu\text{F}} \cdot C_{\text{DM},i}[\mu\text{F}] + 1.54 \text{ cm}^3. \quad (31)$$

This is a fit to the boxed volumes computed for the capacitors with a lead spacing of 27.5 mm given in [64], which are considered to be most suitable for the investigated output filter.

⁶Relation (30) is experimentally derived generating only winding losses in the components. However, as stated in [63, Remark 2, Sec. 2], the same relation was found while heating the core with only core losses.

B. ρ

m

The loss and volume models are employed to compute the ρ - η PF of the two-stage LC filter [cf., Fig. 2(b)] based on the 4D DS with design variables $L_{\text{DM},1}$, $C_{\text{DM},1}$, n , and k (detailed in Section III-G). Fig. 7 depicts the flowchart of the optimization procedure. In a first step, the different criteria for the CPS derived in Section III are evaluated for high numbers of values of $L_{\text{DM},1}$, $C_{\text{DM},1}$, n , and k (box ① in Fig. 7). In order to achieve a constant relative (percentaged) change between adjacent values of inductances and capacitances, i.e., $L_{\text{DM},1,i+1}/L_{\text{DM},1,i} = \text{const.}$ and $C_{\text{DM},1,j+1}/C_{\text{DM},1,j} = \text{const.}$, geometric series are selected

$$\begin{aligned} L_{\text{DM},1,i} &= L_{\text{DM},1,\text{min}} \cdot 10^{\frac{i}{48}}, \\ L_{\text{DM},1,\text{min}} &= 100 \mu\text{H}, 0 \leq i \leq 32, i \in \mathbb{N}_0 \\ C_{\text{DM},1,j} &= C_{\text{DM},1,\text{min}} \cdot 10^{\frac{j}{12}}, \\ C_{\text{DM},1,\text{min}} &= 1 \mu\text{F}, 0 \leq j \leq 16, j \in \mathbb{N}_0. \end{aligned} \quad (32)$$

TABLE III
TWO-STAGE LC OUTPUT FILTER [CF., FIG. 2(B)] DESIGN RESULTING FROM THE ρ - η PARETO OPTIMIZATION (CF., FIGS. 7 AND 9) AND ITS HARDWARE REALIZATION (CF., SECTION IV-A)

τ_m	θ_m	τ_{θ}	θ	θ^*	θ_m
	m	B	θ_m	48 kHz	B
$L_{DM,1}$	154 μ H	146.8 cm^3	$4 \times 2 \times \text{E } 47/20/16$ (ferrite N87, EPCOS) $N = 13, \varnothing_{cu} = 2.5 \text{ mm}, d_{air} = 1.83 \text{ mm}$	154.2 μ H	157.3 cm^3
$C_{DM,1}$	4.6 μ F	20.6 cm^3	4.7 $\mu\text{F}_{rated} 0.68 \mu\text{F}_{rated}$ X2/305V V_{rms} AC B32924C3475M and B32924C3684M (MKP, EPCOS)	4.7 μ F	24.0 cm^3
$L_{DM,2}$	11.6 μ H	9.4 cm^3	$3 \times 2 \times \text{E } 20/10/6$ (ferrite N27, EPCOS) $N = 8, \varnothing_{cu} = 1.4 \text{ mm}, d_{air} = 1.03 \text{ mm}$	11.7 μ H	10.2 cm^3
$C_{DM,2}$	4.1 μ F	18.7 cm^3	4.7 μF_{rated} X2/305V V_{rms} AC B32924C3475M (MKP, EPCOS)	4.1 μ F	17.9 cm^3
$L_{D,2}$	23.1 μ H	16.6 cm^3	$2 \times 2 \times \text{E } 25/13/7$ (ferrite N27, EPCOS) $N = 14, \varnothing_{cu} = 1.8 \text{ mm}, d_{air} = 1.78 \text{ mm}$	22.4 μ H	19.4 cm^3
$R_{D,2}$	1.36 Ω	Neglected	2.67 $\Omega 2.7 \Omega = 1.34 \Omega$ Thick film 2512 SMD 1W resistors	1.34 Ω	0.01 cm^3

* Measured with the Agilent impedance analyzer 4294A (40 Hz — 110 MHz) and without pre-magnetization for the inductors and DC voltage offset for the capacitors.

ρ - η plane; cf., Fig. 8(a) for $L_{DM,1}$ [⑤ and ⑥ in Fig. 7].⁹ An inductor design with ρ_j and η_j is on the PF if its η_j is greater than the efficiencies η of all other designs with a higher power density $\rho > \rho_j$.

To find filter realizations with minimal volume ($\Rightarrow$ maximum power density) for a given efficiency and/or with minimum losses ($\Rightarrow$ maximum efficiency) for a given volume, only the inductor designs on the corresponding inductor PF can be considered, because the inductor designs for given inductance values $L_{DM,1}$, $L_{DM,2}$, and $L_{D,2}$ are independent from each other, and the losses and volumes are strictly greater than zero, i.e. $P_{L_{DM,1}} > 0$, $P_{L_{DM,2}} > 0$, and $P_{L_{D,2}} > 0$ and $V_{L_{DM,1}} > 0$, $V_{L_{DM,2}} > 0$, and $V_{L_{D,2}} > 0$. Therefore, only 46 designs on the inductor PF are considered for $L_{DM,1}$, 51 for $L_{DM,2}$, and 51 for $L_{D,2}$, resulting in $46 \times 51 \times 51 = 119646$ two-stage filter realizations [gray points in Fig. 8(b)]. The power density ρ_q and efficiency η_q of a filter design q can be computed according to

$$\rho_q = \frac{P_{A,out,n}}{V_{L_{DM,1},q} + V_{C_{DM,1},q} + V_{L_{DM,2},q} + V_{C_{DM,2},q} + V_{L_{D,2},q}}$$

$$\eta_q = \frac{P_{A,out,n}}{P_{A,out,n} + (P_{L_{DM,1},q} + P_{L_{DM,2},q} + P_{L_{D,2},q} + P_{R_{D,2},q})} \quad (35)$$

where $P_{A,out,n}$ is equal to 10 kW/3 and $P_{R_{D,2},q}$ denote the losses of the damping resistor $R_{D,2}$. To further reduce the number of filter realizations of the selected point in the DS ($L_{DM,1} = 205 \mu\text{H}$, $C_{DM,1} = 6.5 \mu\text{F}$, $n = 0.076$, and $k = 0.9$), again, only the resulting filter ρ - η PF as depicted in Fig. 8(b), consisting of 559 points, is taken and stored [⑦ in Fig. 7].

As explained for the selected quadruple [cf., (34)], the filter PF for each quadruple in the DS is computed in the same manner, and the final ρ - η PF of the two-stage LC output filter, as shown in Fig. 9 (black points), can be calculated from the filter PFs [⑧

⁹Because the voltage across $C_{DM,1}$ is assumed to be constant, the inductor PFs for all inductance values of $L_{DM,1}$ are only computed once and stored in a lookup table (cf., Fig. 7).

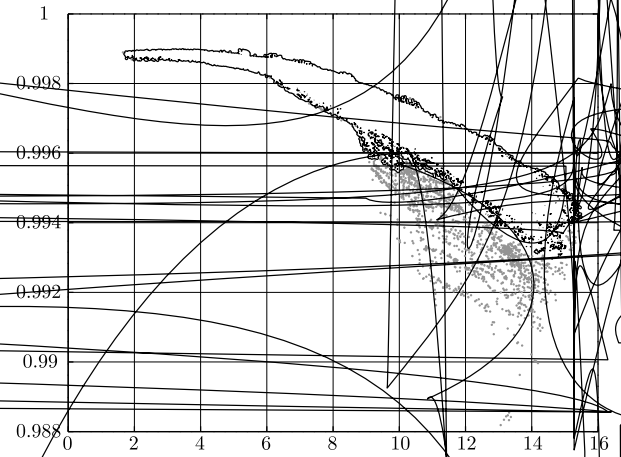


TABLE IV
COMPARISON BETWEEN THE CALCULATED AND MEASURED PERFORMANCE INDEXES AS DEFINED IN SECTION III (CF., TABLE II) FOR THE ρ - η PARETO OPTIMIZED AND REALIZED TWO-STAGE LC OUTPUT FILTER [CF., FIG. 2(b)] WITH THE MEASURED VALUES AS GIVEN IN TABLE III

m	ρ	η	ρ	η
Output voltage slew rate (cf., Section III-A)	min. 203 V/ms	322 V/ms	324 V/ms	
Transient output voltage dip $\rightarrow Z_{step}$ (cf., Section III-B)	max. 5.6 Ω	4.7 Ω	4.2 Ω	
Bridge-leg current ripple Δi_{A0} (cf., Section III-C)	max. 12.3 A	12.0 A	12.2 A	
Output voltage ripple $\Delta v_{A,out}$ (cf., Section III-D)	max. 22.8 V	2.5 V	2.6 V	
Capacitive reactive power Q_{cap} (cf., Section III-E)	max. 333 VA	147 VA	146 VA*	
Conducted EMI limit (cf., Section III-F)	max. 79 dB μ V ($f < 500$ kHz)	62.4 dB μ V @ 192 kHz	57.1 dB μ V @ 192 kHz	

* This value is calculated with $(230 \text{ V})^2 / |Z_{out}|$ based on the measured output impedance $Z_{out} = 363 \Omega \cdot e^{-j \cdot 89.97^\circ}$ at $f_{out} = 50$ Hz with the input of the filter being open.

this context, it should again be noted that a margin of 15 dB is included in the filter design to ensure a very low EMI noise level of the generated output voltage.

As can be seen by comparing columns three and six of Table IV, the boxed volumes of the built inductors $L_{DM,1}$, $L_{DM,2}$, and $L_{D,2}$ are 7%, 8%, and 14% larger than the calculated ones, resulting in a measured power density of $\rho_{meas} = 14.6 \text{ kW/dm}^3$ (239 W/in³) instead of $\rho = 15.7 \text{ kW/dm}^3$ (257 W/in³) [deviation of 7%]. The main reason is that each turn, with the wires of the selected wire diameters, requires a certain radius of curvature to be wound around the middle leg of the E-core.

To measure accurately the total losses of the output filter for the conditions given in (28), i.e., $I_{A,out} = 17 \text{ A}$ and $V_{A,out} = 200 \text{ V}$ for $V_{dc} = V_{dc,max} = 800 \text{ V}$, resulting in a modulation index of $m = 0.5$ and/or the maximum peak-to-peak bridge-leg output current ripple, the DC losses were measured separately from the AC losses. DC losses of 13.0 W were measured in the thermal equilibrium for a DC current $I_{A0} = I_{A,out} = 17 \text{ A}$. The measured value results from the multiplication of the measured DC voltage and DC current at the input of the filter for a shorted output. The voltage and the current were measured with Multimeters Fluke 189, which leads to an error of 1% [66] for the DC power (loss) measurement. The calculated DC losses of 12.8 W agree with the measured value (deviation of 2%). On the other hand, AC losses of 5.3 W were measured with a Yokogawa WT3000 power analyzer for no load, $V_{A,out} = 200 \text{ V}$, and $V_{dc} = V_{dc,max} = 800 \text{ V}$, and just after the entire filter was operated in the thermal equilibrium under the conditions of (28), in order to measure the losses for component temperatures corresponding to the mentioned conditions. According to the data sheet of the power analyzer [67], the error of this measurement is 11%. A comparison to the calculated value of 6.6 W reveals that the calculation overestimates the AC losses by 20%. Accordingly, the total measured losses of the filter are $13.0 \text{ W} + 5.3 \text{ W} = 18.3 \text{ W}$ (measurement error of $\pm 4\%$), which is in good agreement with the calculated losses of $12.8 \text{ W} + 6.6 \text{ W} = 19.4 \text{ W}$ (deviation of 6%). Thus, the calculated and measured efficiencies are both 99.4%.

Moreover, the measured (black line) and calculated (gray lines) input to output transfer function ($v_{A,out}/v_{A0}$) of the realized two-stage LC filter with series RL damping branch of the second stage, is depicted in Fig. 10, showing a good agreement

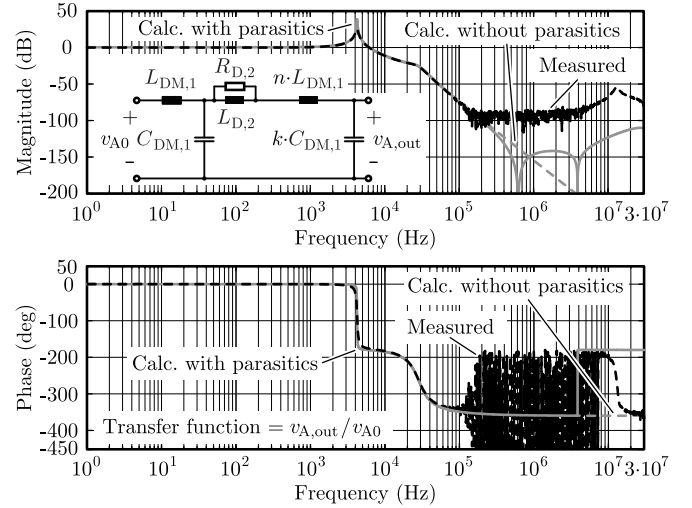


Fig. 10. Measured (black dashed line) and calculated (gray lines) input to output transfer function of the realized two-stage LC filter with a series RL damping branch of the second stage (cf., Table III). It is remarked that the first resonance peak at 4.2 kHz is actively attenuated by the current control loop [cf., Fig. 2(b)] and that the second resonance peak at 27 kHz is passively damped by the serial RL branch. For the dashed gray line ideal components are assumed, whereas for the solid gray line the parasitic capacitances and inductances of the inductors and capacitors (determined by separate measurements) are included. The measurement was conducted with a Bode 100 network analyzer (Omicron Lab, [68]) and shows a good agreement with the calculation until 100 kHz, where the noise floor of the measurement device is reached (specified as -100 dB in [68]).

until 100 kHz, where the noise floor of the measurement device is reached for attenuations higher than 100 dB.

It is noted that the maximum inductor temperatures T_{meas} were only roughly measured as summarized in Table V for the conditions as employed in the calculation given by (28), i.e., $I_{A,out} = 17 \text{ A}$ and $V_{A,out} = 200 \text{ V}$ for $V_{dc} = V_{dc,max} = 800 \text{ V}$. The ambient temperature was $T_a = 30^\circ \text{C}$. The relative errors $(T_{meas} - T_a)/(T_{calc} - T_a)$ of the measurements compared to the calculated temperatures T_{calc} are -16% , 27% , and 14% for $L_{DM,1}$, $L_{DM,2}$, and $L_{D,2}$, respectively, and are in the expected tolerance range. The measured (over) temperatures for $L_{DM,2}$ and $L_{D,2}$ could be explained by the close proximity of $L_{DM,2}$ and $L_{D,2}$.

In the following, the measurement results for each requirement are discussed. It is anticipated that for the measurement of the output voltage slew rate, transient output voltage dip,

(

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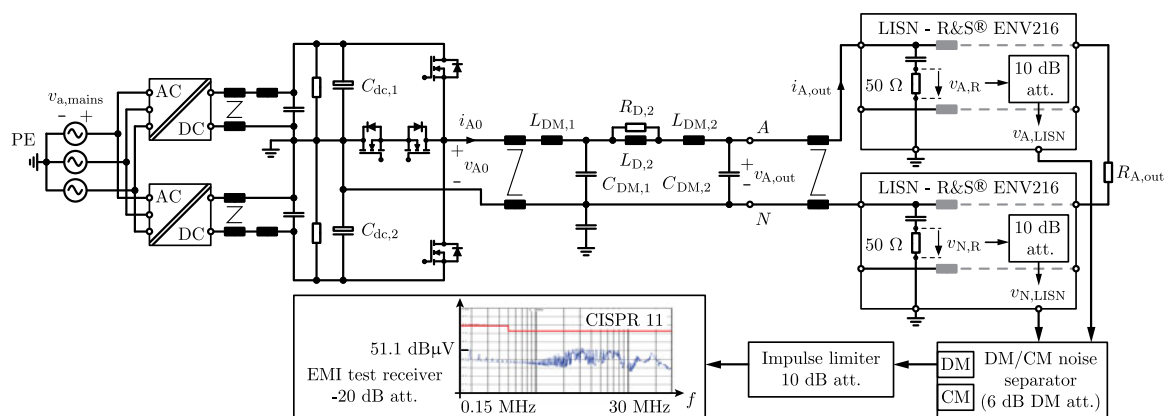


Fig. 15. Conducted EMI measurement setup to measure the DM noise emission of the CPS employing two LISNs at the filter output and a DM/CM noise separator. The attenuations of 10 dB of the LISNs (for increased accuracy) and of 10 dB of the impulse limiter (used to protect the test receiver) are compensated by the -20 dB of the EMI test receiver. The additional attenuation of 6 dB of the noise separator is counterbalanced by adding 6 dB to the measurement results a posteriori (cf., Fig. 16). Thus, in (41) the actually generated noise level is given.

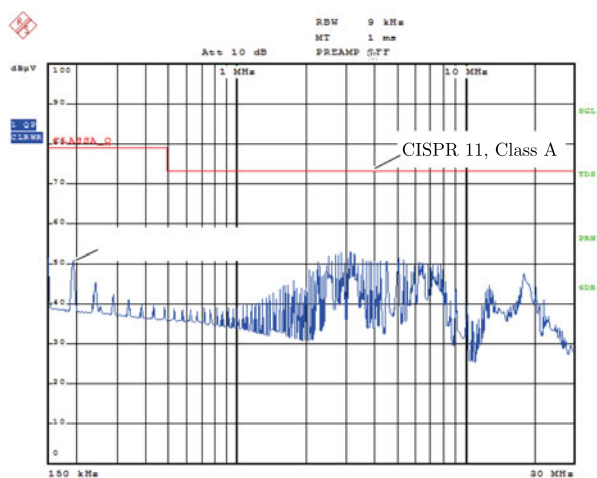


TABLE VI

BEST CASE, NOMINAL CASE, AND WORST CASE PERFORMANCE INDEXES CALCULATED FOR THE COMPONENT VALUES OF THE FINAL FILTER DESIGN, I.E., $L_{DM,1} = 154 \mu\text{H}$, $C_{DM,1} = 4.7 \mu\text{F}$, $L_{DM,2} = 11.7 \mu\text{H}$, $C_{DM,2} = 4.1 \mu\text{F}$, $L_{D,2} = 22.4 \mu\text{H}$, AND $R_{D,2} = 1.34 \Omega$ (CF., TABLE III), AND FOR INDUCTANCE TOLERANCES OF $\pm 10\%$ AND CAPACITANCE TOLERANCES OF $\pm 20\%$

m	B	m	θ	W	W	θ_m	D
Out. voltage slew rate	360 V/ms	322 V/ms	275 V/ms	$L_{DM,1} = 154 \mu\text{H} + 10\%$, $C_{DM,1} = 4.7 \mu\text{F} + 20\%$	$L_{DM,2} = 11.7 \mu\text{H} + 10\%$, $C_{DM,2} = 4.1 \mu\text{F} + 20\%$	$L_{D,2} = 22.4 \mu\text{H} + 10\%$, $R_{D,2} = 1.34 \Omega$	
Trans. out. voltage dip $\rightarrow Z_{step}$	4.1 Ω	4.7 Ω	5.4 Ω	$L_{DM,1} = 154 \mu\text{H} + 10\%$, $C_{DM,1} = 4.7 \mu\text{F} - 20\%$	$L_{DM,2} = 11.7 \mu\text{H} + 10\%$, $C_{DM,2} = 4.1 \mu\text{F} - 20\%$	$L_{D,2} = 22.4 \mu\text{H} + 10\%$, $R_{D,2} = 1.34 \Omega$	
Bridge-leg current ripple Δi_{A0}	10.8 A	12.0 A	13.4 A	$L_{DM,1} = 154 \mu\text{H} - 10\%$, $C_{DM,1} = 4.7 \mu\text{F} - 20\%$	$L_{DM,2} = 11.7 \mu\text{H} - 10\%$, $C_{DM,2} = 4.1 \mu\text{F} - 20\%$	$L_{D,2} = 22.4 \mu\text{H} - 10\%$, $R_{D,2} = 1.34 \Omega$	
Out. voltage ripple $\Delta v_{A,out}$	1.4 V	2.5 V	5.6 V	$L_{DM,1} = 154 \mu\text{H} - 10\%$, $C_{DM,1} = 4.7 \mu\text{F} - 20\%$	$L_{DM,2} = 11.7 \mu\text{H} - 10\%$, $C_{DM,2} = 4.1 \mu\text{F} - 20\%$	$L_{D,2} = 22.4 \mu\text{H} + 10\%$, $R_{D,2} = 1.34 \Omega$	
Cap. reactive power Q_{cap}	117 VA	147 VA	175 VA	$C_{DM,1} = 4.7 \mu\text{F} + 20\%$, $C_{DM,2} = 4.1 \mu\text{F} + 20\%$			
Conducted EMI limit	60.4 dB μ V	62.4 dB μ V	71.1 dB μ V	$L_{DM,1} = 154 \mu\text{H} - 10\%$, $C_{DM,1} = 4.7 \mu\text{F} - 20\%$	$L_{DM,2} = 11.7 \mu\text{H} - 10\%$, $C_{DM,2} = 4.1 \mu\text{F} - 20\%$	$L_{D,2} = 22.4 \mu\text{H} + 10\%$, $R_{D,2} = 1.34 \Omega$	

The last column summarizes the worst case component values. The tolerance of $R_{D,2}$ is neglected.

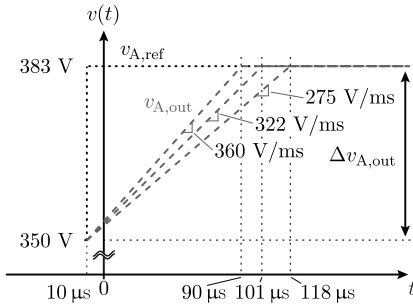


Fig. 17. Best case, nominal case, and worst case slew rates, i.e., 360 V/ms, 322 V/ms, and 275 V/ms, respectively, obtained for the component values of the final filter design considering tolerances of $\pm 10\%$ for the inductances and $\pm 20\%$ for the capacitances. The tolerance of the damping resistor $R_{D,2}$ is neglected. It is noted that for the hardware realization of the filter the components were selected carefully such that the calculated and measured values fit closely as shown in Table III. This minimizes the impact of component tolerances on the output filter performance leading to a good agreement between measured and calculated performance indexes (cf., Table IV).

finally be noted that the resulting filter design is also advantageous concerning output voltage control. Because the resonant frequency of the first filter stage is by a factor of roughly 7 lower than the one of the second filter stage, the second filter stage can be omitted in a first step, i.e., the controller design can be carried out assuming a single-stage LC filter. The closed control loop will, therefore, show a limited gain at frequencies higher than the corner frequency of the first filter stage. Accordingly, the second filter stage cannot be dynamically compensated, which requires to add the passive RL damping.

APPENDIX

DS INCLUDING COMPONENT TOLERANCES

The values of the passive components of the investigated two-stage LC output filter, in particular inductances and capacitances, are subject to tolerances. The impacts of these tolerances on the resulting DS are disregarded in Sections III and IV, to focus on the derivation of the DS and the subsequent ρ - η Pareto optimization.¹² This appendix describes a method, which con-

¹²The hardware realization of the two-stage LC output filter (cf., Table III) uses carefully selected components, such that calculated and measured values agree. For this, the air gaps of the inductors were tuned to achieve the calculated

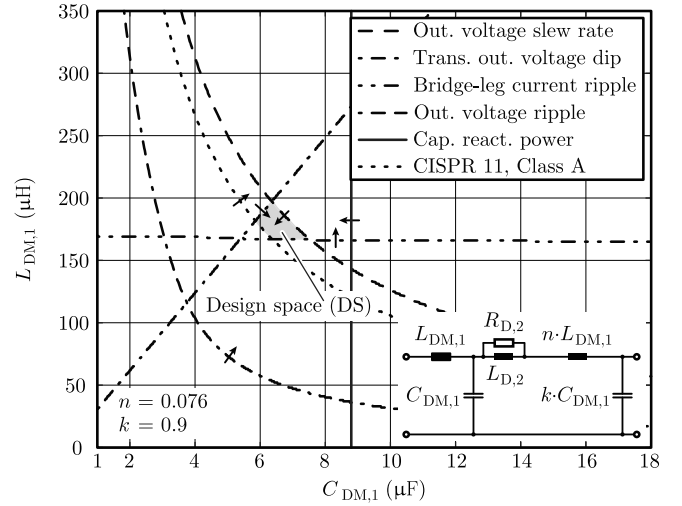


Fig. 18. DS for the two-stage LC output filter [cf., Fig. 2(b)] for nominal $n = L_{DM,2}/L_{DM,1} = 0.076$ and $k = C_{DM,2}/C_{DM,1} = 0.9$, which results from the optimization in Section IV-B (cf., Table III) for the worst case combinations of component values, according to Table VI, if tolerances of $\pm 10\%$ are considered for the inductances and $\pm 20\%$ for the capacitances ($R_{D,2}$ has no tolerance). The resulting DS is considerably smaller than the DS calculated for the exact component values; cf., Fig. 6.

siders the implications of tolerances of the component values on the resulting DS.

To guarantee that all the required properties of the considered AC source specified in Table II can be fulfilled for real components with tolerances, the calculation of the DS needs to include the variation of the component values, because higher or lower than nominal (or desired) component values affect the filter performance indexes listed in Table IV, as exemplary shown in Fig. 17 for the slew rate. The inductors and capacitors are assumed to have tolerances of $\pm 10\%$ and $\pm 20\%$, respectively,¹³ while the tolerance of the damping resistor $R_{D,2}$ is neglected. Thus, the two-stage output filter contains five components with

inductances and the capacitors with the best fitting capacitances were selected from a set of available capacitors.

¹³The inductor is realized with an air gap (cf., Table III) and hence its inductance is roughly proportional to the length of the air gap. It is assumed that the air gap length is 1 mm which can be adjusted within ± 0.1 mm, i.e., within $\pm 10\%$, and hence the tolerance of the inductance is $\pm 10\%$. The tolerance of the capacitance is according to [64] $\pm 20\%$.

uncertain component values. With these component tolerances, the best and worst case performance indexes, e.g., the highest and the lowest slew rates, respectively, can be determined based on given filter component values. This is exemplarily illustrated in Fig. 17 for the nominal filter values identified in the course of this study, i.e., $L_{DM,1} = 154 \mu\text{H}$, $C_{DM,1} = 4.7 \mu\text{F}$, $L_{DM,2} = 11.7 \mu\text{H}$, $C_{DM,2} = 4.1 \mu\text{F}$, $L_{D,2} = 22.4 \mu\text{H}$, and $R_{D,2} = 1.34 \Omega$ (cf., Section IV-B and Table III). In this example, the worst case is obtained for the values of $L_{DM,1}$, $L_{DM,2}$, $L_{D,2}$, $C_{DM,1}$, and $C_{DM,2}$ being 10% and 20% higher, respectively.

Table VI summarizes the best and worst case performance indexes calculated for the above mentioned component tolerances and all criteria considered in Section III. The last column of Table VI lists the particular combinations of component tolerances, which lead to the respective worst case performance indexes. According to the results listed in this column, different combinations of component tolerances result for the different criteria.

Fig. 18 depicts the DS that results if each criteria is evaluated based on the respective combinations of component tolerances listed in Table VI, to ensure that all the required properties of the AC source can be met in any case. Compared to the DS calculated with the nominal (exact) component values shown in Fig. 6, the DS with component tolerances as given in Fig. 18 is considerably smaller.

REFERENCES

- [1] M. Oettmeier, R. Bartelt, C. Heising, V. Staudt, A. Steimel, S. B. Tietmeier, Bock, and C. Doerlemann, "Power-electronic-based machine emulator for high-power high-frequency drive converter test," in *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 1–6, 2010.
- [2] M. Oettmeier, R. Bartelt, C. Heising, V. Staudt, A. Steimel, S. Tietmeier, B. Bock, and C. Doerlemann, "Machine emulator: Power-electronics based test equipment for testing high-power drive converters," in *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 582–588, 2010.
- [3] Y. Srinivasa Rao and M. C. Chandorkar, "Real-time electrical load emulator using optimal feedback control technique," *IEEE Trans. Ind. Appl.*, vol. 57, no. 4, pp. 1217–1225, Apr. 2010.
- [4] G. Lauss, F. Lehmann, A. Viehweider, and T. Strasser, "Power hardware in the loop simulation with feedback current filtering for electric systems," in *IEEE Trans. Ind. Appl.*, vol. 47, no. 3, pp. 3725–3730, 2011.
- [5] N. Kim, S.-Y. Kim, H.-G. Lee, C. Hwang, G.-H. Kim, H.-R. Seo, M. Park, and I.-K. Yu, "Design of a grid-simulator for a transient analysis of grid-connected renewable energy system," in *IEEE Trans. Ind. Appl.*, vol. 46, no. 3, pp. 633–637, 2010.
- [6] R. Lohde and F. W. Fuchs, "Laboratory type PWM grid emulator for generating disturbed voltages for testing grid connected devices," in *IEEE Trans. Ind. Appl.*, vol. 45, no. 1, pp. 1–9, 2009.
- [7] S. Turner, D. J. Atkinson, A. G. Jack, and M. Armstrong, "Development of a high bandwidth multi-phase multilevel power supply for electricity supply network emulation," in *IEEE Trans. Ind. Appl.*, vol. 41, no. 1, pp. P.1–P.7, 2005.
- [8] P. J. Tritschler, E. Rullière, and S. Bacha, "Emulation of fuel cell systems," in *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 1–5, 2010.
- [9] R. Zhang, M. Cardinal, P. Szczesny, and M. Dame, "A grid simulator with control of single-phase power converters in d-q rotating frame," in *IEEE Trans. Ind. Appl.*, vol. 38, no. 3, pp. 1431–1436, 2002.
- [10] W. Ren, M. Steurer, and L. Qi, "Evaluating dynamic performance of modern electric drives via power-hardware-in-the-loop simulation," in *IEEE Trans. Ind. Appl.*, vol. 44, no. 3, pp. 2201–2206, 2008.
- [11] C. Heising, R. Bartelt, M. Oettmeier, V. Staudt, and A. Steimel, "Analysis of single-phase 50-kW 16.7-Hz PI-controlled four-quadrant line-side converter under different grid characteristics," *IEEE Trans. Ind. Appl.*, vol. 57, no. 2, pp. 523–531, Feb. 2010.
- [12] (2014, Apr. 29). Spitzenberger & Spies GmbH & Co. [Online]. Available: <http://www.spitzenberger.de/Produkte.aspx>
- [13] (2014, Apr. 29). Ametek Programmable Power. [Online]. Available: <http://www.calinst.com/products/index.htm>
- [14] D. O. Boillat and J. W. Kolar, "Integrated isolation and voltage balancing link of 3-phase 3-level PWM rectifier and inverter systems," in *IEEE Trans. Ind. Appl.*, vol. 40, no. 3, pp. 1073–1080, 2004.
- [15] IEEE Std 1547-2003 (R2008), 2003 (Reaffirmed 2008).
- [16] IEC Standard CISPR 11, Edition 5.0, May 2009.
- [17] D. O. Boillat, J. W. Kolar, and J. Mühlethaler, "Volume minimization of the main DM/CM EMI filter stage of a bidirectional three-phase three-level PWM rectifier system," in *IEEE Trans. Ind. Appl.*, vol. 49, no. 3, pp. 2008–2019, 2013.
- [18] T. Friedli, M. Hartmann, and J. W. Kolar, "The essence of three-phase PFC rectifier systems—Part II," *IEEE Trans. Ind. Appl.*, vol. 29, no. 2, pp. 543–560, Feb. 2014.
- [19] Y. Levron, H. Kim, and R. Erickson, "Design of EMI filters having low harmonic distortion in high-power-factor converters," *IEEE Trans. Ind. Appl.*, vol. 29, no. 7, pp. 3403–3413, Jul. 2014.
- [20] S. B. Dewan and P. D. Ziogas, "Optimum filter design for a single-phase solid-state UPS system," *IEEE Trans. Ind. Appl.*, vol. 1A-15, no. 6, pp. 664–669, Nov. 1979.
- [21] L. Michels, R. F. De Camargo, F. Botteron, H. A. Grudling, and H. Pinheiro, "Generalised design methodology of second-order filters for voltage-source inverters with space-vector modulation," *IEEE Trans. Ind. Appl.*, vol. 153, no. 2, pp. 219–226, 2006.
- [22] J. Muehlethaler, M. Schweizer, R. Blattmann, J. W. Kolar, and A. Ecklebe, "Optimal design of LCL harmonic filters for three-phase PFC rectifiers," *IEEE Trans. Ind. Appl.*, vol. 28, no. 7, pp. 3114–3125, Jul. 2013.
- [23] Z. Zou, Z. Wang, and M. Cheng, "Modeling, analysis, and design of multifunction grid-interfaced inverters with output LCL filter," *IEEE Trans. Ind. Appl.*, vol. 29, no. 7, pp. 3830–3839, Jul. 2014.
- [24] T.-F. Wu, L.-C. Lin, N. Yao, Y.-K. Chen, and Y.-C. Chang, "Extended application of D-Σ digital control to single-phase bi-directional inverter with LCL filter," *IEEE Trans. Ind. Appl.*, vol. 99, pp. 1–10, 2014, to be published.
- [25] A. Bjelečić, L. Corradini, D. Maksimovic, and R. Zane, "Specifications-driven design space boundaries for point-of-load converters," in *IEEE Trans. Ind. Appl.*, vol. 47, no. 3, pp. 1166–1173, 2011.
- [26] C. Ó Mathúna, J. A. Cobos, B. Allard, and N. Wang, "Power supply on chip," in *IEEE Trans. Ind. Appl.*, vol. 29, no. 1, pp. 250–259, 2014, Tutorial, p. 250.
- [27] J. W. Kolar, J. Biela, S. Waffler, T. Friedli, and U. Badstuebner, "Performance trends and limitations of power electronic systems," in *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 1–20, 2010.
- [28] P. Lefranc, X. Jannot, and P. Dessante, "Virtual prototyping and pre-sizing methodology for buck DC-DC converters using genetic algorithms," *IEEE Trans. Ind. Appl.*, vol. 5, no. 1, pp. 41–52, 2012.
- [29] A. Husain and K.-Y. Kim, "Design optimization of manifold microchannel heat sink through evolutionary algorithm coupled with surrogate model," *IEEE Trans. Ind. Appl.*, vol. 3, no. 4, pp. 617–624, Apr. 2013.
- [30] P. K. Rout, D. P. Acharya, and G. Panda, "A multiobjective optimization based fast and robust design methodology for low power and low phase noise current starved VCO," *IEEE Trans. Ind. Appl.*, vol. 27, no. 1, pp. 43–50, Feb. 2014.
- [31] H. Mukaidani and H. Xu, "Pareto optimal strategy for stochastic weakly coupled large scale systems with state dependent system noise," *IEEE Trans. Ind. Appl.*, vol. 54, no. 9, pp. 2244–2250, Sep. 2009.
- [32] S. X. Chen and H. B. Gooi, "Jump and shift method for multi-objective optimization," *IEEE Trans. Ind. Appl.*, vol. 58, no. 10, pp. 4538–4548, Oct. 2011.
- [33] Q. Li, M. Liu, and H. Liu, "Piecewise normalized normal constraint method applied to minimization of voltage deviation and active power loss in an AC-DC hybrid power system," *IEEE Trans. Ind. Appl.*, vol. 99, pp. 1–9, Aug. 2014.
- [34] C. Marxgüt, J. Muehlethaler, F. Krismer, and J. W. Kolar, "Multiobjective optimization of ultraflat magnetic components with PCB-integrated core," *IEEE Trans. Ind. Appl.*, vol. 28, no. 7, pp. 3591–3602, Jul. 2013.
- [35] T. M. Andersen, C. M. Zingerli, F. Krismer, J. W. Kolar, N. Wang, and C. O. Mathúna, "Modeling and pareto optimization of microfabricated inductors for power supply on chip," *IEEE Trans. Ind. Appl.*, vol. 28, no. 9, pp. 4422–4430, Sep. 2013.

- [36] X.-B. Hu, M. Wang, and E. Di Paolo, "Calculating complete and exact Pareto front for multiobjective optimization: a new deterministic approach for discrete problems," *IEEE Trans. Evol. Comput.*, vol. 43, no. 3, pp. 1088–1101, Jun. 2013.
- [37] A. De Nardo, N. Femia, G. Petrone, and G. Spagnuolo, "Optimal buck converter output filter design for point-of-load applications," *IEEE Trans. Power Electron.*, vol. 57, no. 4, pp. 1330–1341, Apr. 2010.
- [38] J. Ackermann, "Parameter space design of robust control systems," *IEEE Trans. Automat. Contr.*, vol. 25, no. 6, pp. 1058–1072, Dec. 1980.
- [39] V. Besson and A. T. Shenton, "An interactive parameter space method for robust performance in mixed sensitivity problems," *IEEE Trans. Automat. Contr.*, vol. 44, no. 6, pp. 1272–1276, Jun. 1999.
- [40] M. D. Lutovac, D. V. Tošić, and B. L. Evans, "Design space approach to advanced filter design," in *Proc. IEEE Conf. Decis. Control*, 1997, pp. 179–190.
- [41] C. Fernandez, A. Lazaro, P. Zumel, V. Valdivia, C. Martinez, and A. Barrado, "Design space boundaries of linear compensators applying the k-factor method," in *Proc. IEEE Conf. Decis. Control*, Mar. 2013, pp. 2706–2711.
- [42] R. Dorrance, F. Ren, Y. Toriyama, A. A. Hafez, C.-K. K. Yang, and D. Markovic, "Scalability and design-space analysis of a 1T-1MTJ memory cell for STT-RAMs," *IEEE Trans. Electron. Devices*, vol. 59, no. 4, pp. 878–887, Apr. 2012.
- [43] P. Cortés, D. O. Boillat, H. Ertl, and J. W. Kolar, "Comparative evaluation of multi-loop control schemes for a high-bandwidth AC power source with a two-stage LC output filter," in *Proc. IEEE Conf. Decis. Control*, 2012, pp. 1–10.
- [44] T. Nussbaumer, M. L. Heldwein, G. Gong, S. D. Round, and J. W. Kolar, "Comparison of prediction techniques to compensate time delays caused by digital control of a three-phase buck-type PWM rectifier system," *IEEE Trans. Power Electron.*, vol. 55, no. 2, pp. 791–799, Feb. 2008.
- [45] D. O. Boillat, T. Friedli, J. Mühlethaler, J. W. Kolar, and W. Hribernik, "Analysis of the design space of single-stage and two-stage LC output filters of switch-mode AC power sources," in *Proc. IEEE Conf. Decis. Control*, 2012, pp. 1–8.
- [46] M. Schweizer and J. W. Kolar, "Design and implementation of a highly efficient three-level T-type converter for low-voltage applications," *IEEE Trans. Power Electron.*, vol. 28, no. 2, pp. 899–907, Feb. 2013.
- [47] H. Zumbahlen, Ed., *B*. Norwood, MA, USA: Analog Devices, Inc., 2007, p. 1275.
- [48] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *m*, 5th ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 2006, p. 910.
- [49] M. L. Heldwein, T. Nussbaumer, and J. W. Kolar, "Differential mode EMC input filter design for three-phase AC-DC-AC sparse matrix PWM converters," in *Proc. IEEE Conf. Decis. Control*, 2004, vol. 1, pp. 284–291.
- [50] M. Hartmann, H. Ertl, and J. W. Kolar, "EMI filter design for a 1 MHz, 10 kW three-phase/level PWM rectifier," *IEEE Trans. Power Electron.*, vol. 26, no. 4, pp. 1192–1204, Apr. 2011.
- [51] T. Nussbaumer, K. Raggl, and J. W. Kolar, "Design guidelines for interleaved single-phase boost PFC circuits," *IEEE Trans. Power Electron.*, vol. 56, no. 7, pp. 2559–2573, Jul. 2009.
- [52] K. Raggl, T. Nussbaumer, G. Doerig, J. Biela, and J. W. Kolar, "Comprehensive design and optimization of a high-power-density single-phase boost PFC," *IEEE Trans. Power Electron.*, vol. 56, no. 7, pp. 2574–2587, Jul. 2009.
- [53] D. O. Boillat and J. W. Kolar, "Modeling and experimental analysis of a coupling inductor employed in a high performance AC power source," in *Proc. IEEE Conf. Decis. Control*, 2012, pp. 1–18.
- [54] R. W. Erickson, "Optimal single resistor damping of input filters," in *Proc. IEEE Conf. Decis. Control*, 1999, vol. 2, pp. 1073–1079.
- [55] U. Tietze and C. Schenk, *000-000*, 12th ed. Berlin, Germany: Springer-Verlag, 2002.
- [56] A. Van den Bossche and V. C. Valchev, *m*. Boca Raton, FL, USA: CRC Press, 2005.
- [57] J. Mühlethaler, "Modeling and multi-objective optimization of inductive power components," Ph.D. dissertation, ETH Zurich, Zurich, Switzerland, 2012.
- [58] K. Venkatachalam, C. R. Sullivan, T. Abdallah, and H. Tacca, "Accurate prediction of ferrite core loss with nonsinusoidal waveforms using only steinmetz parameters," in *IEEE Trans. Power Electron.*, 2002, pp. 36–41.
- [59] EPCOS AG, "Ferrite and accessories—SIFERRIT material N27," Tech. Rep., Sep. 2006.
- [60] EPCOS AG, "Ferrite and accessories—SIFERRIT material N87," Tech. Rep., Sep. 2006.
- [61] TDK EPCOS. (2014). E/ELP cores and accessories. [Online]. Available: <http://www.epcos.com/epcos-en/529420/products/product-catalog/ferrites-and-accessories/e-elp-cores-and-accessories>
- [62] *0*, IEC Standard IEC 60038, Ed. 7.0, 2009.
- [63] S. A. Mulder, "On the design of low profile high frequency transformers," in *5*, 1990, pp. 141–159.
- [64] EPCOS AG, "Film capacitors—EMI suppression capacitors (MKP)," Series/Type: B32921C/D ... B32928C/D, Jan. 2014.
- [65] *m*, IEC Standard IEC 60063, Ed. 2.0, 1963.
- [66] Fluke Corporation, "Fluke 187/189 true-rms digital multimeter—Extended specifications," Tech. Rep. Rev. A-8/00, 2000.
- [67] Yokogawa Electric Corporation, "WT3000 precision power analyzer," Tech. Rep. Bull. 7603-00E, 2004.
- [68] Omicron Lab, "Bode 100 vector network analyzer technical data," Tech. Rep., 2006.
- [69] P. Cortés, D. O. Boillat, T. Friedli, M. Schweizer, J. W. Kolar, J. Rodriguez, and W. Hribernik, "Comparative evaluation of control schemes for a high bandwidth three-phase AC source," in *7*, 2012, vol. 1, pp. 321–329.
- [70] S. Wang, F. C. Lee, and W. G. Odendaal, "Characterization, evaluation, and design of noise separator for conducted EMI noise diagnosis," *IEEE Trans. Power Electron.*, vol. 20, no. 4, pp. 974–982, Jul. 2005.



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